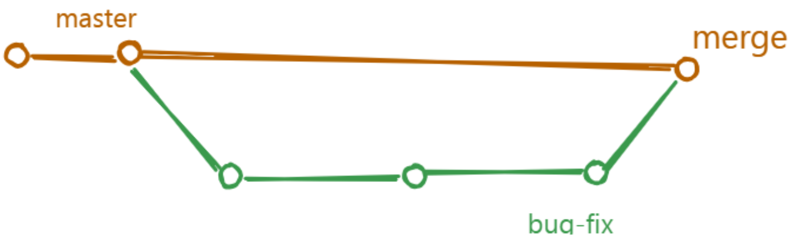
git merge command

-> The git merge command is used to merge one branch into another.
-> We have two types of merge.
   1) Fast-forward merge
   2) Not Fast-forward merge

1) Fast-forward merge

-> It is simple and easy.
-> In Fast forword merge our child branch is ahed of parent.
-> Git simply move the current branch pointer forward to the latest commit

Consider below diagram



-> If there no conflict then we can merge one branch into another branch.

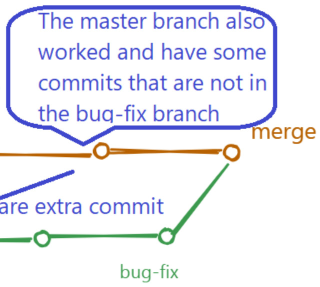
-> 2) Not fast-forward merge

In this type of merge, the parent branch also worked and have some commits that are not child branch. This is a not fast-forward merge.

In a fast-forward merge, there are no conflicts. But in a not fast-forward merge, there are conflicts, and there are no shortcuts to resolve them. You have to manually resolve the conflicts.

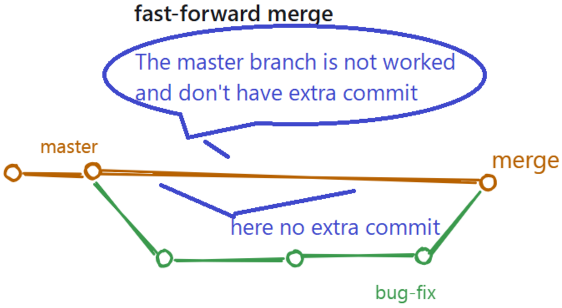
Consider Below Diagram

Not fast-forward merge



Master Branch Pointer is also moved

fast-forward merge



Master Branch is not moved

Q. What is mean by conflict?

-> Conflict means mis-match in the content.

Q. Why conflict came while merging one branch?

-> Conflict means mis-match match in the content.

-> In General, we create a child branch and add our changes in that branch and finally we merge child branch into parent branch.While merging a branch git compare old version of file with new version and git expect old version changes should be there in addition to new changes.If old version changes are not there then we will get conflict.

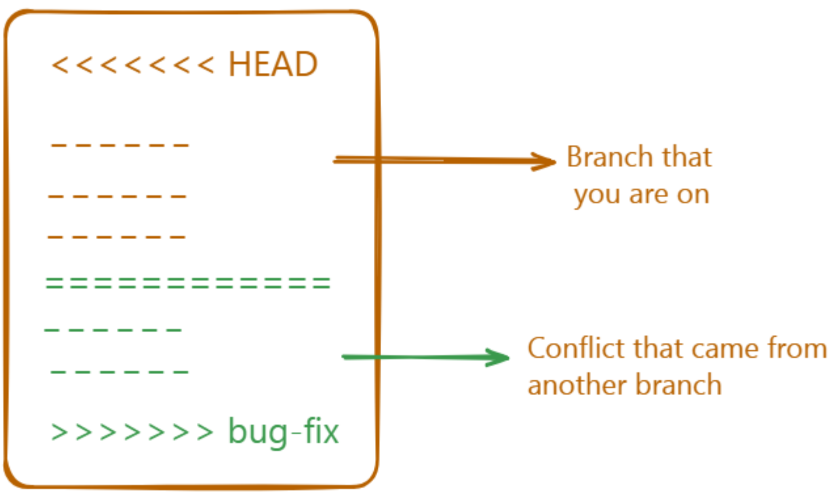
Q. How to Resolve the conflict?

-> There are no shortcuts to resolve conflict. You have to resolve manually.

-> Steps to resolve the conflict

In Local repository, merge other branch changes into your branch by using git merge other_branch_name

One we execute above command then other branch changes get added into your branch and your files look like



-> While resolveing the conflict you have to decide what to keep and what to discard. modify the file accordingly, then commit the changes to local repository and finally push on remote repository and try to merge your branch into parent branch.

-> Ones merging is completed then delete your child branch.

-> Now, your remote repository parent branch(main/release) is updated but your local repository parent branch(main/release) is not in sync with remote repository. Now we need to make in sync.

-> To make local repository main branch in sync with remote repository main branch, perform the following action.

-> switch to main branch : git checkout main

-> execute git fetch oring main command

-> execute git pull command.

Now everything is upto date and in sync.

Q. How to work on git in real Time?

Suppose you have assigned one task, to complete the task, your have to follow the below steps.

Step 1: Clone a repository if not available on your system.
git clone repository_url.

Step 2: Create a branch on github either from main or another branch.

Step 3: Checkout(clone) that branch in the local repository.
Before cloneing branch check all branches by using
"git branch -a"
It will list out local + remote branches.
If newly created branch name is not reflecting in the list
then use "git fetch" then clone your branch

git checkout -b branch_name origin/branch-name

Step 4: Add your changes and commit your changes.

Step 5: push your local repository changes to remote repository by using
git push

Step 6: Raise pull request to merge your branch into parent branch.

Steps to Raise Pull Request

1. Click on "Compare and Pull Request"
2. Compare Source and Target Branch
3. Add description
4. Add Assignee
5. Create Pull request
6. Share pull request URL to Approver via mail
7. Ones approved then merge it and close pull request
8. Delete your branch.